

Root-Locus Plots with MATLAB

Drawing a locus and reading gains and closed-loop poles off it.

Ogata, *Modern Control Engineering*, Ch. 6.

In this tutorial you will:

- plot a locus with `rlocus` and find closed-loop poles for given gains,
- check the asymptotes and breakaway point by hand, and
- find the gain at the imaginary-axis crossing (the stability limit).

Step through with **Ctrl+Enter**, or render a report with `publish`.

This file demonstrates the `rlocus` command on a representative open-loop transfer function and shows how to read closed-loop pole locations and the corresponding gain directly from the plot.

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Open-Loop System

$$G(s) = \frac{K}{s(s+1)(s+3)}$$

```
num = 1;
den = conv([1 1 0],[1 3]);
G = tf(num,den);
poles_ol = pole(G)
zeros_ol = zero(G)

figure
rlocus(G)
title('Root Locus: $G(s)=\frac{1}{s(s+1)(s+3)}$', 'Interpreter', 'latex', 'FontSize', 20)
grid on

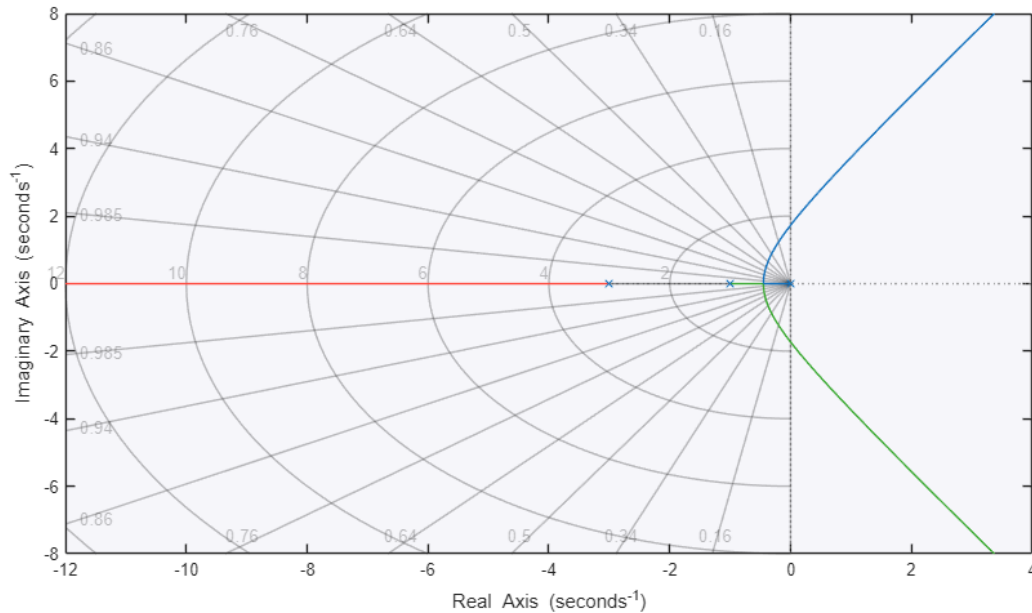
poles_ol =

    0
   -3
   -1

zeros_ol =

0×1 empty double column vector
```

$$\text{Root Locus: } G(s) = \frac{1}{s(s+1)(s+3)}$$



Reading Gain and Closed-Loop Poles at a Chosen Point

`rlocfind` (interactive) or, programmatically, `rlocus` with a specific gain vector can be used to find the closed-loop poles for a given K . Here we sweep a vector of gains and report the resulting pole locations directly.

```
Ks = [0.5 1 2 4 8];
for K = Ks
    cl_poles = pole(feedback(K*G,1));
    fprintf('K = %4.1f -> closed-loop poles: %s\n', K, mat2str(cl_poles,4))
end
```

```
K = 0.5 -> closed-loop poles: [-3.078;-0.6846;-0.2373]
K = 1.0 -> closed-loop poles: [-3.148;-0.4261+0.369i;-0.4261-0.369i]
K = 2.0 -> closed-loop poles: [-3.27;-0.3652+0.6916i;-0.3652-0.6916i]
K = 4.0 -> closed-loop poles: [-3.468;-0.2662+1.041i;-0.2662-1.041i]
K = 8.0 -> closed-loop poles: [-3.767;-0.1163+1.453i;-0.1163-1.453i]
```

Asymptotes and Breakaway Point (Hand Calculation Check)

With three poles at 0, -1, -3 and no finite zeros ($n = 3, m = 0$):

$$\sigma_a = \frac{(0)+(-1)+(-3)}{3} = -\frac{4}{3}$$

$$\theta_a = 60^\circ, 180^\circ, 300^\circ$$

```
sigma_a = sum(poles_ol)/numel(poles_ol);
fprintf('Asymptote centroid sigma_a = %.4f (expect -1.3333)\n', sigma_a)
```

```
% The breakaway point satisfies dK/ds = 0 for K = -1/G(s); solve
% numerically with the characteristic polynomial's derivative.
```

```
syms s
Gs = 1/(s*(s+1)*(s+3));
Ksym = -1/Gs;
dKds = diff(Ksym,s);
breakaway_candidates = double(solve(dKds==0,s));
disp('Breakaway candidates (real, between 0 and -1):')
disp(breakaway_candidates(imag(breakaway_candidates)==0))
```

```

Asymptote centroid sigma_a = -1.3333 (expect -1.3333)
Breakaway candidates (real, between 0 and -1):
-2.2153
-0.4514

```

Gain at the Imaginary-Axis Crossing (Stability Boundary)

The locus crosses into the right half-plane at some gain K_{cr} ; this is found from the Routh array of the closed-loop characteristic polynomial $s^3 + 4s^2 + 3s + K = 0$ exactly as in `Transient and Steady-State/RouthStability.m`. Routh: s^3 : [1 3], s^2 : [4 K], s^1 : [(12-K)/4], s^0 : [K] Stability requires $0 < K < 12$.

```

K_cr = 12;
p_cr = roots([1 4 3 K_cr]);
disp('Closed-loop poles at the stability boundary K=12:')
disp(p_cr)

```

```

Closed-loop poles at the stability boundary K=12:
-4.0000 + 0.0000i
-0.0000 + 1.7321i
-0.0000 - 1.7321i

```

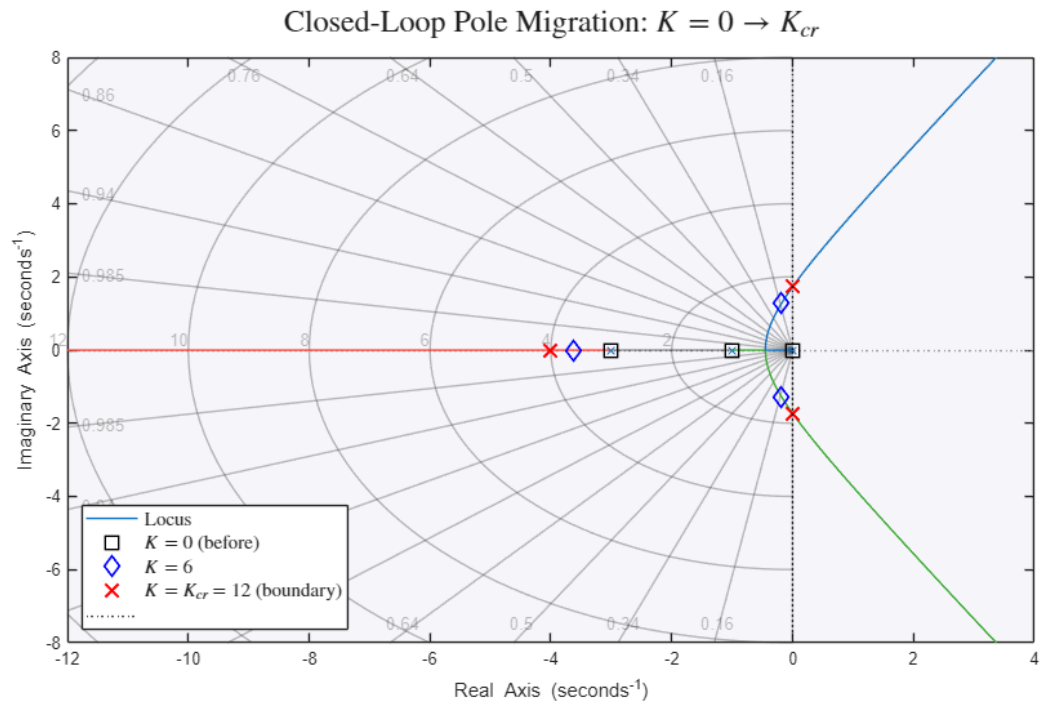
Before vs. After: How the Poles Migrate with Gain

The locus **is** the before/after story. At $K=0$ (before) the closed-loop poles sit exactly on the open-loop poles; as K grows they migrate along the branches, reaching the imaginary axis at $K = K_{cr}$ (onset of instability). Overlay the $K=0$ starting points, an intermediate K , and the $K=K_{cr}$ endpoints on the locus.

```

p0 = poles_ol; % K = 0 (before)
pmid = pole(feedback(6*G,1)); % K = 6 (intermediate)
% p_cr already holds the K = Kcr boundary poles
figure
rlocus(G)
hold on
plot(real(p0),imag(p0),'ks','MarkerSize',10,'LineWidth',1.5)
plot(real(pmid),imag(pmid),'bd','MarkerSize',8,'LineWidth',1.5)
plot(real(p_cr),imag(p_cr),'rx','MarkerSize',11,'LineWidth',1.5)
xline(0,'k:')
hold off
legend('Locus','$K=0$ (before)', '$K=6$', '$K=K_{cr}=12$ (boundary)', ...
'Interpreter','latex','FontSize',11,'Location','southwest')
title('Closed-Loop Pole Migration: $K=0 \rightarrow K_{cr}$','Interpreter','latex','FontSize',18)
grid on % keep rlocus's own "Real/Imaginary Axis" labels (it re-appends units,
% which breaks a custom latex xlabel/ylabel)

```



Try it yourself

- Sweep a finer gain vector and find where the branches cross the imaginary axis – that gain matches the Routh limit $K=12$.
- Add a zero at -2 ($G = \text{tf}([1 \ 2], \text{den})$) and watch a branch terminate on it instead of escaping to infinity.